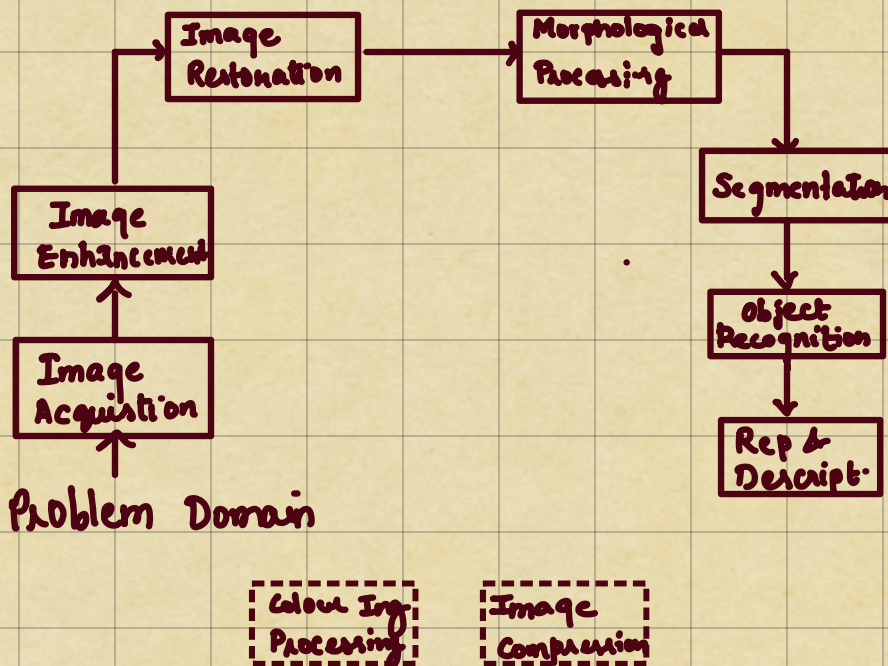


What is Digital Image Processing ?

- ① Improvement of pictorial info for human interpretation.
- ② Processing image data for storage, transmission and representation for autonomous machines perception

Light : $\lambda = c/f$

Key stages in Digital Image Processing



light on obj $\rightarrow$ capt by sensor $\rightarrow$ volt. wf

Img capt. = Analog $\xrightarrow{\text{SP}}$ cont.
To discretize we use sampling & quantization

- ① Image Acquisition - retrieving images from some source as per the problem domain
- ② Image Enhancement - improving quality of img for future processing ↓ subj step
- ③ Image Restoration - restore the img affected by noise. ← obj.
- ④ Colour Image Processing - deals with colour img. instead of gray.
- ⑤ Compression - techniques req. for reducing the storage req. to save an img & b/w req. to transmit
- ⑥ Morphological - trying to understand the structure of img. extract the img components
- ⑦ Segmentation - dividing img into meaningful objects / regions
- ⑧ Object Recognition - identify the obj. is based on description
- ⑨ Rep & Description - after segmentation represent objects mathematically

Intensity Monochrome - each pix $\rightarrow$ light intensity on grayscale
 $\rightarrow$ RGB / colour $\rightarrow$ each pixel contains a vector $\Rightarrow$ RGB
Image Types $\rightarrow$ Black-white / Binary $\rightarrow$ 1 = white, 0 = black.

Intensity Resolution / Effect of Quant. levels

$\downarrow$
Grey scale = $\overset{\downarrow \text{black}}{0} - 255 \leftarrow \text{white}$
= 256 levels $\rightarrow$ high detail

Quantization = reducing no. of diff. intensity levels

256 = v. smooth 128 = similar 64 = less smooth

False Contours = smooth entirely $\rightarrow$ visible steps

$\downarrow$ quantization

$$I = 2^k$$

$\swarrow$ no. of bits / pix
 $\nwarrow$ intensity levels

Representation of Pixels

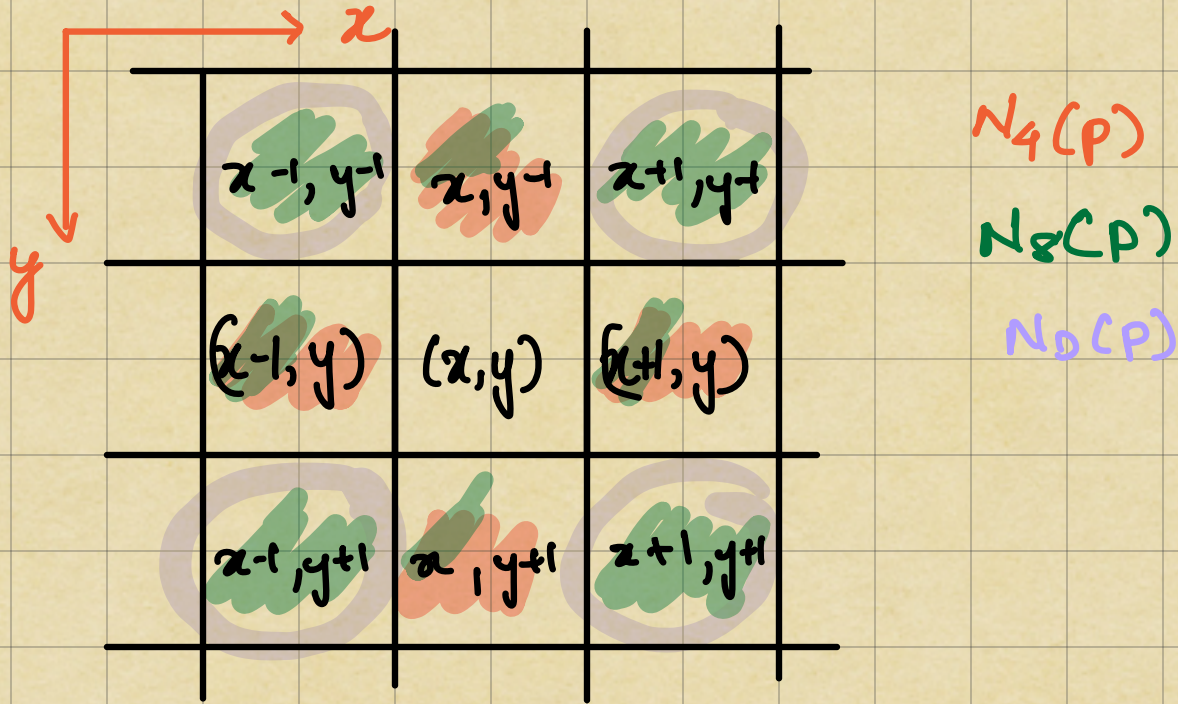
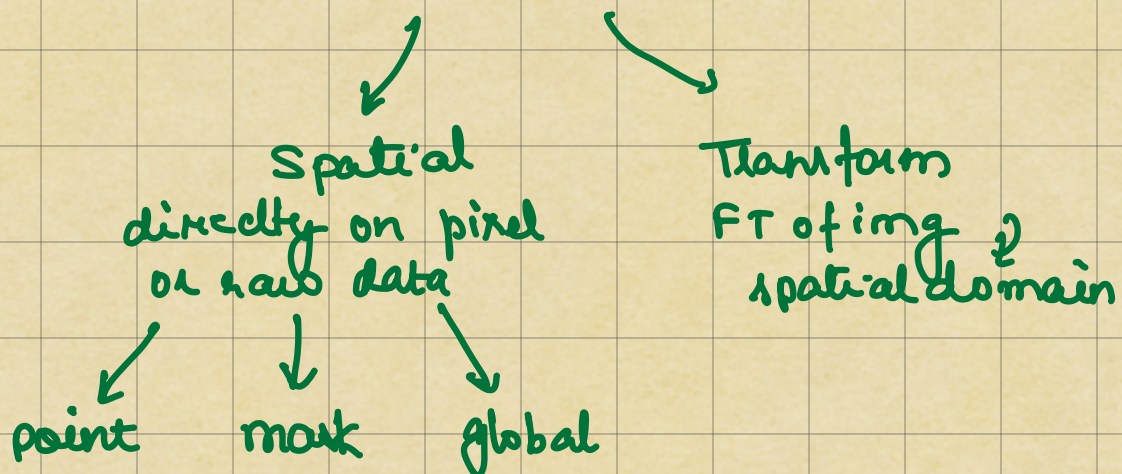


Image Enhancement

- improve the interpretability of the info in the img for human viewers
- enhancement algos yield better quality images with the purpose of diff appn
 - image contrast
 - noise suppressing



Point Operation : $g(m,n) = T[f(m,n)]$

one to one mapping b/w i/p & o/p img

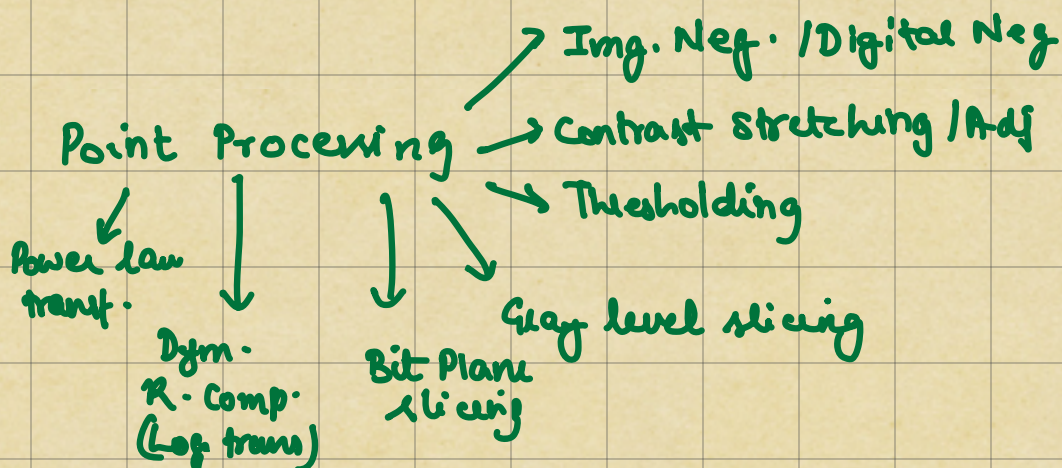
Mask Operation : Transform T operates on a neighbourhood of pixels.

Mask a small matrix with values $\Rightarrow$ weights

$\hookrightarrow$ origin sym $\rightarrow$ centre asym $\rightarrow$ anywhere

Global Operation : apply to all pixel values in the img

$\hookrightarrow$ freq. domain



1. Image Negative

$$F = \text{img}$$

black $\downarrow$ 0-255 $\uparrow$ white

$F(x,y) \rightarrow \text{pixel}$

$\Rightarrow$ Invert grey level / reverse light & dark
 $\Rightarrow$ X-rays

$$g(m,n) = 255 - f(m,n)$$

$\uparrow$
(-1)

8 bits/p = 256 levels. $\therefore 0-255$

$$\therefore 255 - f(m,n)$$

Q. 8 BPP img.

obtain image -ve

$$g(m,n) = 255 - f(m,n)$$

121	205	217	156	151
139	127	157	117	125
252	117	236	138	142
227	182	172	197	242
201	106	119	251	240

$$\frac{g(m,n)}{255 - f(m,n)}$$

134	50	38	99	104
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

1. Contrast Stretching / Adj.

- poor illumination img
- dark → darker bright → brighter

$$g(m,n) = k * f(m,n)$$

↑
constant

$k > 1$ ↑ contrast

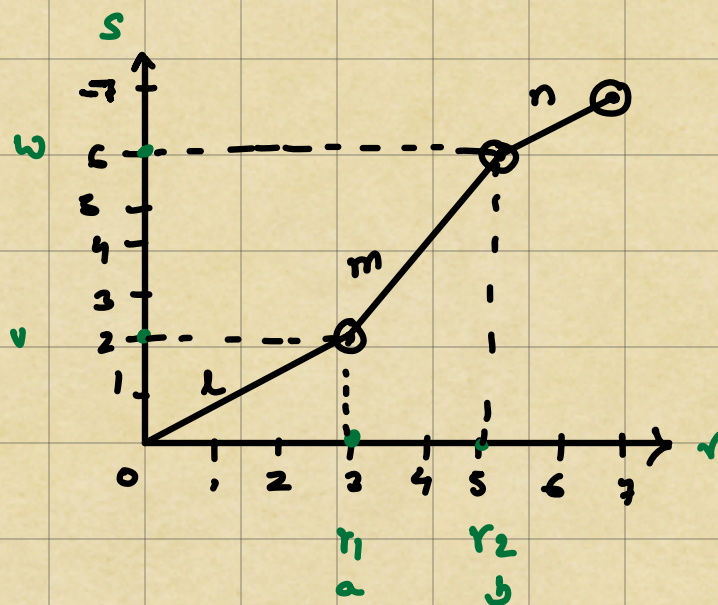
$k < 1$ ↓ "

Q. $L=8$

3	1	2	4
5	1	6	2
2	3	5	6

$$r_1 = 3 \quad r_2 = 5$$

$$s_1 = 2 \quad s_2 = 6$$



$$L = \frac{2-0}{3-0} = 0.66$$

$$m = \frac{6-2}{5-3} = 2$$

$$n = \frac{7-6}{7-5} = 0.5$$

$$S = \begin{cases} L * r & 0 \leq r < a \\ m * (r-a) + v & a \leq r < b \\ n * (r-b) + w & b \leq r < L-1 \end{cases} \rightarrow S = \begin{cases} 0.66 * r & 0 \leq r < 3 \\ - & 3 \leq r < 5 \\ - & 5 \leq r < 7 \end{cases}$$

calce. s for present r then substituted ans.
 val.

3. Thresholding

Extreme contrast stretching

$$s = 255$$

$$= 0$$

if $r \geq a$ threshold

$r < a$

Q.

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

$$a = 4$$

$$2^3 = 8$$

$0 \rightarrow 7$

$$s = 7$$

$$= 0$$

$$r \geq 4$$

$$r < 4$$

7 0 7 0
 0 7 7 7
 0 0 7 7
 7 7 7 0

Q.

1	0	6	11
8	15	7	10
10	4	7	9
12	5	13	2

$$2^4 = 16$$

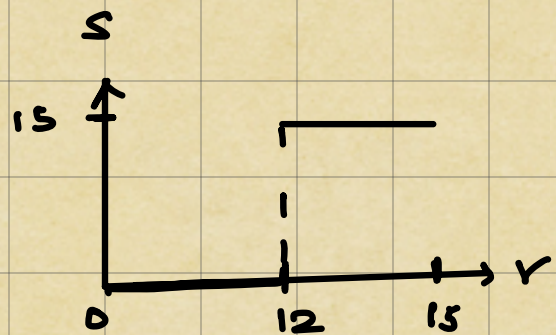
0-15

$$T = 12$$

$$S = \begin{cases} 15 & r \geq 12 \\ 0 & r < 12 \end{cases}$$

→

0	0	0	0
0	15	0	0
0	0	0	0
15	0	15	0



4. Gray Lvl Slicing $\begin{matrix} \nearrow \text{with bg} \\ \rightarrow \text{w/o bg} \end{matrix}$

$$S = L-1$$

$$s = 0$$

$$S = L-1$$

$$s = r$$

$$r_1 = 3 \quad r_2 = 5 \quad = 345 \quad (L-1)$$

$$= < 3 \quad (0/r)$$

$\rightarrow$ Thresholding splits the grey lvl $\rightarrow$ 2 parts

Q.

5	1	3	2	7
0	3	1	1	6
0	1	4	4	2
1	6	3	2	5
3	3	4	<u>7</u>	5

$$r_1 = 3 \quad r_2 = 5$$

$$2^3 = 8 \leftarrow \text{lvl}$$

Gray lvl slicing

w/o bg

$$L-1 = 7 \quad r_1 \rightarrow r_2$$

$$r \quad < r_1 \quad > r_2$$

w bg

$$L-1$$

$$0 \quad < r_1 \quad > r_2$$

7	1	7	2	7
0	7	1	1	6
0	1	7	7	2
1	6	7	2	7
7	7	7	7	7

7	0	7	0	0
0	7	0	0	0
0	0	7	7	0
0	0	7	0	7
7	7	7	0	7

5. Bit plane slicing

↳ Pixel = number → composed of bits

↳ Instead of highlighting grey-level → contribution made by each bit

↳ Useful in image compression

0 0 1
MSB LSB

⇒ 8 bit image = 8-bit planes

↳ Often by sep. bits → highlight interesting aspects of img
high order = sign. visual info lower order = subtle det.

Q. 1 2 0
 4 3 2
 ⑦ 5 2

$\therefore 8 \text{ levels} = 3 \text{ bits} = 3 \text{ 1-bit planes}$

Binary Image

$\Rightarrow$

001	010	^M 00 ^L 0
100	011	010
111	101	010

Bit-1

LSB

1	0	0
0	1	0
1	1	0

Bit-2

Mid-Bit plane

0	1	0
0	1	1
1	0	1

Bit-3

MSB bitplane

0	0	0
1	0	0
1	1	0

6. Dynamic Range Compression (Log Transf.)

- Dynamic range exceeds the capability of the display device
- Some pixels are so high they overshadow the lower valued pixels
- $\therefore$ Dynamic Range Comp. is used
- log being the best funcn. compressor is used
 $\Rightarrow$ logarithmic transformation

$$S = \underset{\substack{\uparrow \\ \text{constant}}}{C} * (1 + \log |a|) \quad \swarrow \text{image}$$

Q.

110	(120)	90
91	94	98
90	91	99

$$L = 2^7 = (128)$$
$$n = 7$$

(i) $C = 1$

(ii) $C = \frac{L}{\log_{10}(1+L)}$

(i)

n	S	
110	$1^* \log(1+n)$	$= 2.04 \sim 2$
120	$1^* \log(1+120)$	$= 2.08 \sim 2$
90	$1^* \log(1+90)$	$= 1.95 \sim 2$
91	$1^* \log(1+91)$	$= 1.96 \sim 2$
94	$1^* \log(1+94)$	$= 1.97 \sim 2$
98	$1^* \log(1+98)$	$= 1.99 \sim 2$
99	$1^* \log(1+99)$	$= 2 \sim 2$

 $\Rightarrow$

2 2 2

2 2 2

2 2 2

o/p image

(ii)

 $C =$

$$\frac{L}{\log(1+L)}$$

 $=$

$$\frac{128}{\log(129)}$$

60.66

n	S
110	$60.66 * \log(1+110) = 123.7 \sim 124$
120	$60.66 * \log(1+120) = 126.2 \sim 126$
90	$60.66 * \log(1+90) = 118.28 \sim 118$
91	$60.66 * \log(1+91) = 118.89 \sim 119$
94	$60.66 * \log(1+94) = 119.5 \sim 120$
98	$60.66 * \log(1+98) = 120.7 \sim 121$
99	$60.66 * \log(1+99) = 121.32 \sim 121$

124	126	118
119	120	121
90	119	121

7. Power law Transformation

- Humans perception of brightness - more sensitive to changes in dark than to bright
- Display devices do not display images linearly voltage & intensity follow a non-linear power function.
 - ↳ darker than intended images
- So we use Gamma correction ↗ cancels out the effect

$$S = c * r^{\gamma}$$

Q. similar to log transform just change formula

Mask / Neighbourhood Operations

1. Low Pass Averaging Filter

→ used to eliminate Gaussian noise ↗ random variations in intensity

→ the mean filter replaces every pixel with the avg. of the neighbours

* → low-pass filter preserves the smooth region in image & removes the sharp variations producing blurring

3x3 low spatial mask

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

5x5

$$\frac{1}{25} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

→ sum of elements in low pass spatial mask = 1

↳ preserves overall bright of image

→ ↑ mask ⇒ blurring ↑ ⇒ stronger smoothing

↳ odd size

Q.

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

$$mask = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Iteration 1:

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 10 & 10 & 10 & 10 & 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 & 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 & 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 & 10 & 10 & 10 & 10 \\ 50 & 50 & 50 & 50 & 50 & 50 & 50 & 50 \\ 50 & 50 & 50 & 50 & 50 & 50 & 50 & 50 \\ 50 & 50 & 50 & 50 & 50 & 50 & 50 & 50 \\ 50 & 50 & 50 & 50 & 50 & 50 & 50 & 50 \end{bmatrix}$$

$$= \frac{1}{9} (1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10)$$

$$= \underline{\underline{10}}$$

$$\frac{1}{9} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} * \begin{bmatrix} 10 & 10 & 10 & 10 & 10 & 10 & 10 & 10 \\ 10 & & 10 & 10 & 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 & 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 & 10 & 10 & 10 & 10 \\ 50 & & 50 & 50 & 50 & 50 & 50 & 50 \\ 50 & 50 & 50 & 50 & 50 & 50 & 50 & 50 \\ 50 & 50 & 50 & 50 & 50 & 50 & 50 & 50 \\ 50 & 50 & 50 & 50 & 50 & 50 & 50 & 50 \end{bmatrix}$$

$$= \frac{1}{9} (1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 50 + 1 \times 50 + 1 \times 50)$$

$$= 23.33$$

$$= \frac{1}{9} (1 \times 10 + 1 \times 10 + 1 \times 10 + 1 \times 50 + 1 \times 50 + 1 \times 50 + 1 \times 50 + 1 \times 50 + 1 \times 50)$$

$$= 36.67$$

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10

23-33 23-33 23-33 23-33 23-33 23-33 23-33 23-33

36-67 36-67 36-67 36-67 36-67 36-67 36-67 36-67

50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

Limitations

- blurring $\Rightarrow$ affects feature localisation
- corrupted by impulse noise $\Rightarrow$ attenuated & diffused $\Rightarrow$ x x error
- a single pixel with unrepresentative value = affect on centre pixel

2. Weighted Avg

$$\frac{1}{16} \times$$

1	2	1
2	4	2
1	2	1

pixels nearest to the center have ↑ values than others

pixel at center always multiplied by a higher value
⇒ giving more imp. in avg.

3. Median Filter

- non-linear filters in the spatial domain
- smoothing using median of neighbourhood

Steps: 1) All values identified by mask written in asc/desc order
2) median of this list chosen as centre pixel value.

Q. 1 5 7 (2x3) mask

2 (4) 6

3 2 1

Step 1: ~~1~~ ~~1~~ ~~2~~ ~~2~~ 3 ~~4~~ ~~5~~ ~~6~~ ~~7~~

Step 2:
 $\uparrow$
 median = 3

1 5 7

2 (4) 6

3 2 1

→

1 5 7

2 3 6

3 2 1

Q- 18 22 33 25 32 24

34 128 24 172 26 23

22 19 32 31 28 26

18	22	33	25	32	24
34	128	24	172	26	23
22	19	32	31	28	26

→ 128

18 19 22 22 (24) 32 33 34 128

↑ median

→ 24

19 22 24 25 (31) 32 33 128 172

↑ m

→ 172

24 25 26 28 (31) 32 32 33 172

→ 26

23 24 25 26 (26) 28 31 32 172

18	22	33	25	32	24		18	22	33	25	32	24
34	128	24	172	26	23	→	34	24	31	31	26	23
22	19	32	31	28	26		22	19	32	31	28	26

when we take median value, pixel values are v.

diff from their neighbourhood pixels → equal to neigh.

↓

capable of reducing salt & pepper noise

↓
impulse noise → some pixels of
img are $\leq$ max int.
min int.

3. High Pass filter

- eliminates low freq. regions while retaining or enhancing high freq. components
- high pass img = no bg as low freq. region
- image sharpening = highlight fine details
- sum of mask = 0
 - ↳ remove low. freq.

$$\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

eg:

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
100	100	100	100	100	100	100	100
100	100	100	100	100	100	100	100
100	100	100	100	100	100	100	100
100	100	100	100	100	100	100	100

$$= \frac{1}{9} \times ((-1) \times 10 + (-1) \times 10 + (-1) \times 10 + (-1) \times 10 + 8 \times 10 + (-1) \times 10 + (-1) \times 10 + (-1) \times 10 + (-1) \times 10)$$

$$= \frac{1}{9} (-10 - 10 - 10 - 10 + 80 - 10 - 10 - 10 - 10) = 0$$

$$= \frac{1}{9} (-1 \times 10 - 1 \times 10 - 1 \times 10 - 1 \times 10 + 8 \times 10 - 1 \times 10 - 1 \times 100 - 100 - 1 \times 100)$$

$$= \frac{1}{9} (-10 - 10 - 10 - 10 + 80 - 10 - 100 - 100 - 100)$$

$$= -30 \quad \hookrightarrow \text{pixel val} \neq -ve \quad \therefore 0$$

$$= \frac{1}{9} (-1 \times 10 - 1 \times 10 - 1 \times 10 - 1 \times 100 + 8 \times 100 - 1 \times 100 - 1 \times 100 - 1 \times 100 - 1 \times 100)$$

$$= \frac{1}{9} \times 270 = 30$$

$$= \frac{1}{9} (-1 \times 100 - 1 \times 100 - 1 \times 100 - 1 \times 100 + 8 \times 100 - 1 \times 100 - 1 \times 100 - 1 \times 100 - 1 \times 100)$$

$$= 0$$

⇒

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
20	30	20	30	30	30	30	30
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

4. High Boost filter

→ high pass filter → removes l_g

$$\text{high boost} = (A-1) \times f(m,n) + \text{high pass}$$

→ high freq. emphasis filter retains some l_g
to aid interpretation of image

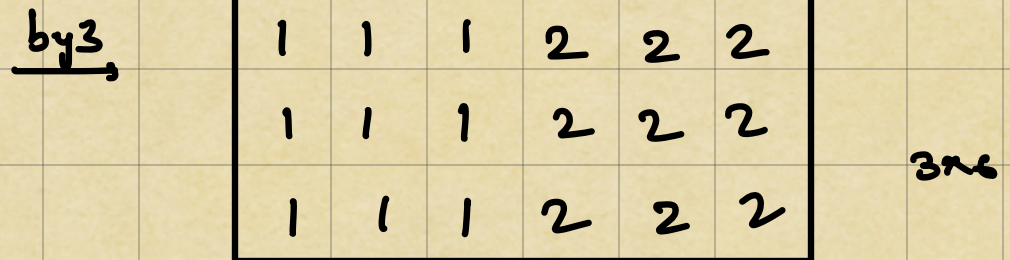
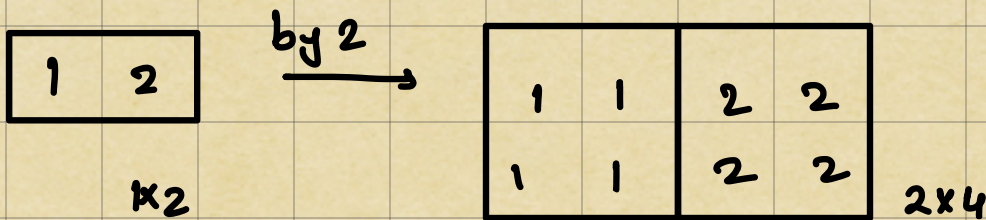
$$\text{HFB} \rightarrow \frac{1}{9} \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 9 & -1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix}$$

5. Zooming

↳ img has fine details may not be able to examine on a std display monitor

↳ helps view fine details

↳ simplest operation of zooming = replication of pixels



Histogram Equalization

- attempts to spread out gray levels in an img so they evenly distributed across their range
- reassign brightness values of pixel based on hist img.
- img = flat

Procedure :

1. Perform running sum
2. Normalize by dividing / total. no of pixels
3. multiply 2 by max-gray lvl and round
4. Map

Q.

4	4	4	4	4
3	4	5	4	3
3	5	5	5	3
3	4	5	4	3
4	4	4	4	4

$2 \times 5 = 8$
 $0 - 7$
 max gray.

Gray line	0	1	2	3	4	5	6	7
No. of pix	0	0	0	6	14	5	0	0
Runn. sum	0	0	0	6	20	25	25	25
Normaliz.	$0/25$	$0/25$	$0/25$	$6/25$	$20/25$	$25/25$	$25/25$	$25/25$
x gray max (x 7) & round	0	0	0	2	6	7	7	7

no. of pix = 25

4	4	4	4	4
3	4	5	4	3
3	5	5	5	3
3	4	5	4	3
4	4	4	4	4



6	6	6	6	6
2	6	7	6	2
2	6	7	6	2
2	6	7	6	2
6	6	6	6	6